$$\vec{\omega}(t) = e^{t}\vec{e_1} + e^{-2t}\vec{e_2} = \begin{bmatrix} e^{t} \\ e^{-2t} \end{bmatrix}$$

∴ Differentiating w.r.t time

$$\dot{\vec{\omega}}(t) = \begin{bmatrix} e^{t} \\ -2e^{-2t} \end{bmatrix}$$

$$\begin{bmatrix} e^{t} \\ -2e^{-2t} \end{bmatrix} = A \begin{bmatrix} e^{t} \\ e^{-2t} \end{bmatrix} \quad -(1)$$

Let $A = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$ $-(2)$

∴ (2) in (1) gives

$$e^{t} = \alpha e^{t} + \beta e^{-2t} \quad -(3)$$

$$-2e^{-2t} = \gamma e^{t} + \delta e^{-2t} \quad -(4)$$

(3) and (4) hold true $\forall t$

$\Rightarrow \alpha = 1 \; ; \; \beta = 0$

$\gamma = 0, \; \delta = -2$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$$